Business Management HL · Set A · Paper 2

Question booklet · 105 minutes · 50 marks

Answer every compulsory question and choose one complete Section B option. A calculator is permitted. Use the case evidence and show calculations. Write on separate paper.

Section A · Polder Panels [12]

Polder manufactures insulation panels. Its latest accounts show current assets of CU 540,000, including inventory of CU 210,000, and current liabilities of CU 300,000. Annual credit sales are CU 1,460,000, and average trade receivables CU 220,000. Use a 365-day year. Its largest customer recently asked for more credit. Polder needs cash to purchase a cutting machine.

(a) Calculate the current ratio and the acid-test ratio. **[4]**

(b) Calculate debtor days. Estimate how much cash would be released if debtor days fell to 40, assuming credit sales were unchanged. **[4]**

(c) Explain two possible consequences of tightening customer credit for Polder. **[4]**

Section A · Tern Works [12]

Tern is deciding whether to automate cutting. A machine costs CU 150,000 now. Net cash inflows are forecast at CU 55,000, CU 65,000 and CU 70,000 in years 1–3. There is no residual value. Assume cash accrues evenly within each year for payback and arrives at year-end for discounting. Use discount factors 0.909, 0.826 and 0.751. Forecast demand depends on renewal of a single contract.

(a) Calculate the payback period in years. **[3]**

(b) Calculate the net present value using the supplied factors. **[4]**

(c) Find the lowest year-3 net cash inflow that gives a non-negative NPV, holding years 1 and 2 unchanged. Explain the significance for Tern. **[3]**

(d) Explain one difference between the information supplied by NPV and by payback. **[2]**

Section A · Fenn Freight [6]

Fenn can lease a depot immediately or run a six-month pilot. All monetary outcomes below are net gains relative to doing nothing. Immediate leasing yields CU 140,000 if demand is high (probability 0.55) and a loss of CU 50,000 otherwise. The pilot yields CU 65,000 or CU 15,000 with the same respective probabilities. Pilot costs are already included.

(a) Using expected monetary values, determine which option is preferred. **[3]**

(b) Find the high-demand probability at which the two options have the same EMV. Explain one reason a manager might still select the pilot at probability 0.55. **[3]**

Section B · Option 1 · Koru Packaging [20]

Koru operates one line with rated capacity of 160,000 cartons per month. Current output is 128,000. Quality rejects are 5% of output. A proposed preventive maintenance programme would reduce capacity to 150,000 and rejects to 2%. Forecast output would then be 135,000. Each saleable carton contributes CU 1.20 before the extra monthly maintenance cost of CU 8,000. A new customer expects at least 130,000 saleable cartons every month and can cancel after two missed deliveries.

(a) Calculate current capacity utilization. **[2]**

(b) Calculate expected saleable output before and after the change and the expected monthly gain in contribution after maintenance costs. **[4]**

(c) Find the highest reject rate that would meet the customer's monthly target at the forecast output. Explain why this threshold matters. **[4]**

(d) Evaluate adopting the preventive maintenance programme. **[10]**

Section B · Option 2 · Reed Rail Kitchens [20]

Reed expects to sell 18,000 meal kits next year. Production requires 0.4 labour hours per kit. Fixed overheads are CU 216,000. Variable materials and labour cost CU 14 per kit. A rail operator offers CU 28 each for 3,000 extra kits. Spare labour capacity is 1,600 hours. The order requires one inspection costing CU 6,000 and no extra fixed overheads. Reed's retail price is CU 34. The customer asks for priority delivery at peak times.

(a) Calculate the budgeted overhead absorption rate per labour hour. **[2]**

(b) Calculate full cost per kit at the budgeted rate and incremental profit from the extra order. **[4]**

(c) Determine the maximum number of regular retail kits that could be displaced before the order ceases to increase profit. Explain the relevance of spare capacity. **[4]**

(d) Discuss whether Reed should accept the rail contract on the terms offered. **[10]**

End of question booklet · 50 marks